What Do Benign-Trained Session Language Models Actually Detect?
A Mechanistic Audit with Corrected Evaluation Labels

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Abstract

We audit language models adapted to non-insider employee logs by separating score construction, evaluation population, input sensitivity, and internal intervention effects. A calendar-label audit corrects the observed CERT r4.2 and r6.2 populations to 1,362 positive days from 70 insiders and 73 days from 5 insiders. The main adapters' training membership is unaffected, but several label-dependent dictionaries, donors, and comparator fits require repair. Corrected main-adapter fold day ROC is 0.538 and 0.583 against unseen users, compared with 0.960 and 0.956 against general-population negatives. In a complete r4.2 score decomposition, averaging behavior targets improves fold day ROC from 0.540 to 0.820 and AP from 0.004859 to 0.025238 while retaining profile context. This is score mitigation, not evidence of internal profile independence. A corrected sparse-feature pilot completes scoring but fails one realized-control check; its diagnostics do not establish attack specificity. We retain the mechanistic method, report the verified results with their scope, and identify the remaining dependencies before reinstating historical causal claims.

1 Introduction

High ranking quality does not identify which information an anomaly detector uses. In employee activity logs, unusual behavior, profile novelty, and training exposure can all influence a score. We ask how these influences can be separated using frozen-model score decompositions, input manipulations, and internal interventions. This recovery edition distinguishes established measurements from claims still awaiting corrected experiments. Its contribution is an explicit audit and evidence record, not a detector leaderboard or a new mathematical theorem.

2 Data, labels, and populations

We study serialized employee activity from CERT r4.2 and r6.2 (Lindauer, 2020; Glasser & Lindauer, 2013; Le et al., 2020). CERT is a synthetic insider-threat benchmark. A record is one user's observed day, assembled from extracted sessions. The `DAY` line contains week and organizational fields, `PSY` contains simulated personality scores, and `SES` lines contain session activity counts. `SESCOUNT` records the number of sessions. These are structured measurements written as text.

Our primary target is membership in the official master's incident window: a day is positive if its recorded calendar interval intersects the user's half-open incident interval $[\text{start}, \text{end})$. Evaluation is restricted to days with extracted sessions. This labels an *incident-window day*, not every action on that day. An insider's other days are called *window-negative*; they are not verified benign activity.

An audit found an incorrect calendar join in the previous evaluation labels. The corrected day-index origin is December 28, 2009. Independent enumeration reproduced every corrected label, and the original texts, ordering, and splits were preserved. The resulting observed populations are:

Dataset	Observed days	Positive days	Insiders	Changed labels
r4.2	330,295	1,362	70	407
r6.2	1,393,297	73	5	23

The r4.2 changes comprise 177 positive-to-negative and 230 negative-to-positive labels; r6.2 has 10 and 13, respectively. The master windows include 530 r4.2 and 26 r6.2 additional days with no extracted session, which are outside this observation domain. Named malicious-event days define a different target; they are retained as sensitivity labels and are not silently substituted for the master-window target. The r6.2 master/detail discrepancies and one repaired malformed master date are documented in the label audit.

Training exposure. The main LM adapters remain usable: their actual training and validation user lists exclude all 70 r4.2 and all 5 r6.2 official insiders. This was checked independently of the faulty day-label join. The main training/validation counts are 851/79 non-insider users for r4.2 and 3,590/405 for r6.2. The unseen-user evaluation pools contain all official insiders plus those validation users: 42,468 days from 149 users and 142,072 days from 410 users. "Unseen" means absent from adaptation; these repeatedly analyzed validation users are not a newly untouched test set. The factorial and smaller-model adapters are separate training runs, with their own provenance qualifications.

Why labels affect more than a metric. Correcting labels can change a feature's rank, the sample used to fit a sparse dictionary, or which user is eligible as a normal donor. In those cases the historical intervention remains a measurement of its original inputs, but it is not the experiment the corrected procedure would have selected. Model training is reusable only when its actual inputs and fitting membership are unchanged. We apply this distinction throughout.

3 What the score measures

For tokens $x_1, \ldots, x_n$, the adapted LM assigns the anomaly score

$$s(x) = -\frac{1}{n-1} \sum_{t=2}^{n} \log p_{\text{ada}}(x_t \mid x_{<t}). \tag{1}$$

The first token is context, padding is excluded, and losses are assigned to the *predicted target token*. A larger score means the observed day was less predictable under the model of the adaptation corpus.

We separate two questions. In the general-population comparison, the negative user pool includes users seen during adaptation. In the unseen-user comparison, negative users are restricted to validation users absent from adaptation. Each fold contains one official insider and a fixed sampled negative-user cohort; metrics are averaged equally over insiders. The general comparison uses 800 non-insider users per fold; the unseen comparison uses all 79 or 405 validation users. An insider's window-negative days are negatives for day metrics. User scores are the maximum over each user's full scored history. The model stays fixed across these evaluation folds.

ROC-AUC measures the probability that a positive example outranks a negative one, with half credit for ties. Average precision (AP) summarizes precision across recall levels. It is not precision at a chosen alert threshold. A fold average and a pooled metric have different weighting and prevalence: neither can be substituted for the other. Unless explicitly labeled pooled, the tables below report equal-insider fold means. Bootstrap intervals resample users and are descriptive conditional on fixed models, settings, and negative cohorts; five r6.2 insiders provide particularly limited uncertainty information.

Changing scored targets. For a target-token class c, let $L_c(x)$ be its summed loss and $N_c(x)$ its number of scored targets. Then

$$s(x) = \frac{\sum_c L_c(x)}{\sum_c N_c(x)}, \qquad s_B(x) = \frac{\sum_{c \in B} L_c(x)}{\sum_{c \in B} N_c(x)}. \tag{2}$$

Our behavior view includes `SESCOUNT`, `SES`, and any residual non-profile target classes; the profile view includes `DAY` and `PSY`. In this score-only comparison the full original input remains visible to the model. A

behavior target can therefore still be predicted from profile information. Removing profile targets from an average, removing profile *input*, and retraining without profile content are three different interventions.

4 Opening the model

The interpretability method remains part of the work. Its purpose is to connect an internal change to a measured output effect, while checking what the edit actually changed. Its historical empirical claims require the repairs listed later; the procedure itself does not depend on a favorable result.

4.1 Adapt the model, then hold it fixed

We adapt Qwen3-8B on non-insider user-days using QLoRA (Qwen Team, 2025; Hu et al., 2022; Dettmers et al., 2023). The base weights are frozen and quantized; small low-rank matrices supply the learned updates. For a projection matrix, the adapted weight is $W' = W + (\alpha/r)BA$. This makes it practical to compare the same model with adapters enabled and disabled. The model is frozen while we audit it. QLoRA supplies an accessible adaptation difference; it does not establish that an LM is the best detector for these data.

4.2 Isolate the adaptation difference

At token position t and layer ℓ, compare the adapted and base states:

$$\delta_t = h_t^{(\ell),\text{ada}} - h_t^{(\ell),\text{base}}. \tag{3}$$

This is the representation difference induced by adaptation on that input. It is not automatically a unique causal decomposition of the network.

4.3 Find a sparse approximation

Standardize each coordinate of δ_t and fit a sparse autoencoder (SAE):

$$q_t = (\delta_t - \mu) \oslash \sigma, \qquad z_t = \text{TopK}(\text{ReLU}(Eq_t + b_e)), \qquad \widehat{q}_t = Dz_t + b_d. \tag{4}$$

Only a few coefficients of z_t are retained. The dictionary columns of D are candidate directions of change, learned to reconstruct the observed deltas (Cunningham et al., 2023; Gao et al., 2024). Sparse approximation is not a proof that these directions are unique concepts. Any cross-dictionary comparison must account for normalization: physical directions are $\sigma \odot D_{:,f}$.

4.4 Select candidates, edit, and measure

An activation difference between incident-window and reference examples can rank candidate features. Selection is a hypothesis-generating step: labels, reference-user eligibility, and the dictionary sample all affect the answer. To test a selected set, edit its coefficients, continue the forward pass, and measure the change in a declared output such as behavior-target NLL. Compare against active features and random directions with the same edited support and matched realized perturbation size. Keep a same-configuration unedited forward pass as the reference. We measure both the requested edit and the change that actually survives low-precision rounding.

A change in output establishes sensitivity to that specified edit. It does not by itself prove that the features encode maliciousness, that the edit improves anomaly ranking, or that the selected set is strictly necessary. Best-of-search edits also differ from an intervention policy chosen before seeing its outcome.

4.5 Locate activation without overinterpreting it

For a fixed set of examples, activation mass at input class c is

$$m_f(c) = \frac{\sum_{x,t:\,\text{class}(t)=c} z_{t,f}(x)}{\sum_{x,t} z_{t,f}(x)}. \tag{5}$$

This describes where a feature activates. A session-position hidden state can use an earlier profile. High profile-position mass in selected features is neither a complete account of the detector nor proof of a profile-only causal mechanism. Corrected attribution must use the corrected selected features and example population. The earlier profile-versus-behavior headline is pending that recovery.

5 Technical specification of the mechanistic audit

5.1 Adaptation and representation

The main adapter configuration uses Qwen3-8B, NF4 four-bit base weights with double quantization, bfloat16 computation, and rank-16 LoRA updates with scaling 32 and dropout 0.05 on attention and MLP projections. The declared adaptation configuration uses maximum length 2,048, effective batch size eight, one epoch, learning rate 1.5×10^{-4}, cosine decay, and 3% warmup. Base and adapted passes use the same input and tokenizer. Training configurations and artifact hashes belong to each run; factorial adapters cannot be substituted for the main adapters merely because their architecture is the same.

Let $\delta_t = h_t^{\text{ada}} - h_t^{\text{base}}$ at audited layer ℓ. With fitted location and scale vectors μ, σ, the SAE minimizes

$$\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \left\| (\delta_t - \mu) \oslash \sigma - \left(D \,\mathrm{TopK}(\mathrm{ReLU}(E q_t + b_e)) + b_d \right) \right\|_2^2, \qquad q_t = (\delta_t - \mu) \oslash \sigma. \tag{6}$$

Both token sampling and optimization randomness must be recorded. The historical SAE seed argument did not seed torch initialization; earlier across-seed changes therefore mix sampling and optimization variability. The corrected fit records both. A dictionary fitted on evaluation examples is transductive even if feature ranking uses separate discovery users. It must not be described as a dictionary learned independently of every evaluation user.

5.2 Selection, attribution, and eligibility

A mean activation-gap screen proposes a feature set,

$$g_f = \mathbb{E}[z_f \mid y = 1, \text{discovery}] - \mathbb{E}[z_f \mid \text{declared reference}], \qquad S = \mathrm{TopFive}(g). \tag{7}$$

The reference population must be explicit: a window-negative insider day is not interchangeable with a non-insider day. Active controls avoid inert features but do not automatically match perturbation size. Token-weighted and equal-user summaries also answer different questions. Attribution mass describes the stated selected dictionary, token positions, and example set; changing any of these requires recomputation. Physical dictionary directions are $v_f = \mathrm{diag}(\sigma) D_{:,f}$, not standardized decoder columns alone.

5.3 Two distinct intervention implementations

Write $\widehat{\delta}_t = \mu + \sigma \odot (D z_t + b_d)$ and choose a modified code z_t'. A residual-preserving additive edit is

$$h_t' = h_t^{\text{ada}} + \sigma \odot D(z_t' - z_t). \tag{8}$$

A reconstruction-replacement edit instead sets

$$h_t' = h_t^{\text{base}} + \mu + \sigma \odot (D z_t' + b_d). \tag{9}$$

Their difference includes the original SAE reconstruction residual. Even if two arms share replacement, downstream nonlinear responses need not cancel that residual's effects. Unchanged coefficients, unedited tokens, and the code bias must be handled consistently with the chosen implementation. Joint edits write the declared coefficients over a declared union of token positions; their interpretation is not a sum of proven single-feature effects.

For an output functional s_V on declared targets V, define $e_S(x) = s_V(x; I_S) - s_V(x; I_0)$, where I_0 is a live, same-configuration reference. A selected-versus-control contrast is $\mathbb{E}_u[\bar{e}_S(u) - \bar{e}_C(u)]$. An attack-specificity

contrast compares this within-user difference across incident-window and matched window-negative receivers. Positive or negative signs must be interpreted using this output definition; an increased NLL is not automatically better anomaly ranking or successful mitigation.

Historical searched-donor repair used

$$\tau = (\overline{\Delta}^{S}_{\text{anomalous}} - \overline{\Delta}^{S}_{\text{noninsider}}) - (\overline{\Delta}^{C}_{\text{anomalous}} - \overline{\Delta}^{C}_{\text{noninsider}}), \tag{10}$$

with the best signed score change retained over searched donors and strengths. That defines a best-of-search estimand. A typical donor effect, a frozen intervention policy, and detector AP after intervention require separate measurements. The historical estimates are not reinstated in this edition.

5.4 Realized perturbations and numerical consistency

Let r_t be the requested float32 update and h_t the live low-precision state. The realized perturbation is

$$\widetilde{r}_t = operatorname{float32}(\text{cast}(h_t + r_t)) - operatorname{float32}(h_t). \tag{11}$$

Matching $\|r_t\|$ does not guarantee matching $\|\widetilde{r}_t\|$. The corrected pilot checks edited support and realized norm per token at a fixed 5% relative tolerance, alongside no-op identity and loss-partition checks. Norm matching is one nuisance control; it does not make random directions semantically equivalent to selected features.

The historical scoring passes differed with batch shape and cross-entropy precision. Their per-example NLL variation, about 0.01 nats in the controlled checks, can exceed the intervention contrasts of interest. The new reference uses batch size one and float32 cross entropy. Cached hidden states are not silently assumed equal to freshly computed states. Before reusing an effect, check adapter, tokenizer/text, dictionary and normalization, feature set, receiver, donor, token support, edit rule, score targets, and numerical configuration. A change in any of these can invalidate exact reuse.

5.5 Frozen cohorts and conditional mathematics

Fixed-input reanalysis retains the originally measured responses and changes only group labels. Post-hoc eligible-partner subsets have a different population from the original sample. Within-user contrasts use users with both corrected classes and do not recover the historical day pairing. All such estimates are exploratory. No coefficient-feasibility statement alone proves output mediation: a code-space target, its physically realizable hidden-state lift, and the model's behavior under that lift are distinct objects. The mathematical branch remains available for later work after these empirical dependencies are repaired.

6 Results supported after label correction

6.1 The comparison population changes the apparent detector quality

Dataset / negatives	Day ROC	Day AP	User ROC
r4.2 / general	0.960	0.011939	0.965
r4.2 / unseen	0.538	0.004978	0.589
r6.2 / general	0.956	0.000585	0.962
r6.2 / unseen	0.583	0.000123	0.649

The main-adapter day ROC changes from 0.960 to 0.538 on r4.2 and from 0.956 to 0.583 on r6.2 when negatives are restricted to unseen users. This is a population/exposure comparison, not a controlled intervention that identifies the sole cause of the gap. In r4.2, nine of the ten insiders newly included by the label repair are scenario-3 users, and their full scores rank incident-window days poorly. Their inclusion materially changes the main result; recovery cannot stop at the previously covered 60 users.

The historical Deep SVDD, recurrent-autoencoder, and Isolation Forest rows are not included as corrected comparisons. Their fitting populations need repair. The corrected LM measurements do not establish a detector leaderboard ordering.

6.2 Behavior-target scoring improves the complete r4.2 evaluation

r4.2 score	Day ROC	Day AP	User ROC
Full score	0.540	0.004859	0.589
Behavior targets	0.820	0.025238	0.825
Profile targets	0.482	0.001353	0.476

This separate, batch-size-one score decomposition covers all 42,468 unseen-pool days, all 1,362 positive days, and all 70 insiders. A 256-example bridge exactly reproduced the existing class-score pass before the missing users were scored. Class losses reconstruct the total score, and an independent check reproduced 840 reported fold metrics to floating-point precision.

With the model and input held fixed, changing the average from all targets to behavior targets raises mean fold day ROC from 0.540 to 0.820. The paired gain is 0.280, with a descriptive 95% user-bootstrap interval [0.204, 0.357]. AP changes from 0.004859 to 0.025238. This supports a *score mitigation* for this model and population. It does not show that profile information has stopped influencing behavior predictions.

Mean fold prevalence is 0.000714; pooled prevalence is 0.032071. The fold AP belongs with the former evaluation construction, not the pooled prevalence. Neither AP value is the percentage of alerts that would be correct at an operational threshold.

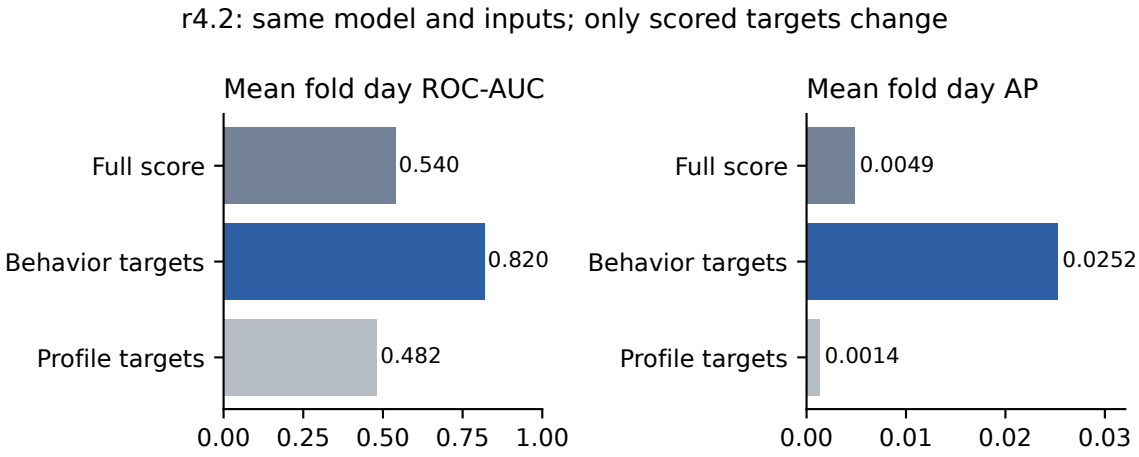


Figure 1: Complete r4.2 score decomposition. All three views retain the same input and frozen model. The figure reports fold means, not pooled metrics.

6.3 Training changes and score changes must stay separate

The next table concerns separate r6.2 factorial adapters, not the main adapters in the first table. Each input condition changes both adaptation and evaluation content, with one adapter seed per condition.

r6.2 adapter / input	Day ROC	User ROC
3B / Full	0.683	0.778
3B / No personality	0.799	0.901
3B / No profile	0.894	0.950
3B / Shuffled profile	0.432	0.550
8B / Full	0.412	0.472
8B / No personality	0.791	0.901
8B / No profile	0.899	0.902
8B / Shuffled profile	0.394	0.438

Removing all profile content improves ranking in both audited model families. The 3B change is substantial in point estimate; the earlier description of a uniformly tiny 3B effect is not supported. These are run-specific comparisons on five insiders, not a general model-capacity law. Their training-byte provenance is weaker than the main-adapter audit and remains explicitly qualified.

Scoring behavior targets while retaining the full input yields another comparison:

Separate r6.2 adapters	Full ROC	Behavior ROC	Behavior AP
3B full training	0.683	0.893	0.005480
3B target-mask training	0.350	0.899	0.003376
8B full training	0.412	0.897	0.001288

The 3B target-mask adapter omits profile targets from its training objective while retaining profile context. Relative to behavior scoring of its full-training counterpart, its day-ROC gain is about 0.006, with a descriptive interval crossing zero, while AP decreases. This does not establish an additional mitigation gain.

Input masking of frozen main adapters is yet another experiment. Removing personality input improves the cached day-ROC point estimates, but removing all profile input does not have the same effect. Those arms used different batch/hardware/cross-entropy configurations; the large personality-removal contrast exceeds the observed metric sensitivity, while small differences cannot be read at fine numerical precision. These findings are compatible with several mechanisms and do not identify one by themselves.

6.4 Temporal models provide a bounded comparison

The corrected cached numeric-model comparison uses a common population of 141,662 history-eligible unseen-user days:

r6.2 numeric model	Pooled day ROC	Pooled AP	User ROC
One observation	0.701	0.001391	0.776
Seven incl. target	0.708	0.002486	0.780
Forecast (past only)	0.726	0.001322	0.645

These are *pooled* metrics. The seven-observation reconstruction includes the target day; the forecast uses preceding observations only. Histories are previous available observations, not necessarily consecutive calendar days. All three checkpoint reloads passed the declared numerical tolerances, although the seven-observation profile score used 99.5% of its tolerance. That is a tolerance pass, not bitwise equality. History-shuffle and broader copying or mechanistic claims remain pending.

7 What the causal evidence currently establishes

7.1 The corrected pilot is an implementation diagnostic

A corrected r4.2 pilot refit the sparse dictionary, reranked features using the corrected discovery labels, and scored 158 incident-window/window-negative pairs from 34 users under 15 conditions. The dictionary uses

layer 26, 8,192 latents, and top-$k = 4$; its sample, fit, and feature selection are recorded. Live batch-size-one reference states and losses were recomputed. All 4,740 condition rows completed and the zero-edit controls reproduced the references.

One dictionary-direction control failed the predeclared 5% realized-norm matching tolerance at one window-negative token (6.42% mismatch). Consequently the original global gate remains failed. The following results are explicitly *post-run diagnostics*, with the original failed gate and all rows retained:

Post-run diagnostic	Effect	95% interval
Selected minus isotropic	+0.005433	[+0.002277, +0.009191]
Selected minus active	+0.004373	[+0.001384, +0.007859]
Attack specificity	-0.003864	[-0.008807, +0.000146]

The first two contrasts show larger behavior-loss increases for selected edits than for their specified controls on incident-window receivers, but the specificity contrast does not establish a larger attack-specific response. These estimates do not reinstate the earlier causal headline. Endpoint-local checks do not retroactively pass the global protocol. A separate control-policy feasibility probe fixes the failing case under a new sampling rule; it is not a replacement full experiment or a theorem of scientific novelty.

7.2 Saved responses retain a narrower meaning

The historical Pilot 2 profile-swap measurements were relabeled on CPU. Of its 414 receivers, 129 change class; 86 have an ineligible partner in at least one scored swap condition. The adapted model remains more sensitive than the base model to the specified profile swaps in the frozen cohort. This measures input sensitivity. Corrected class contrasts and eligible-partner subsets are post-hoc descriptions; they do not restore the original matching or identify harm, mediation, or a complete circuit.

The H6 direct-intervention pilot retains 1,320 scored rows for 120 receivers. Forty-eight receiver labels change, and 44 receivers have at least one official insider among their nominally benign profile partners. Its previously reported small pooled mean effects reproduce. Small means for two specified lifts do not establish equivalence of all feasible implementations or all users. The corrected class summaries are likewise limited to the historical inputs and selected dictionary.

The earlier input-swap experiment retains 723 eligible fixed r4.2 pairs and 60 r6.2 pairs after receiver and donor checks. There are 230 and 13 corrected positive days absent from those original receiver caches, respectively. In r4.2, some other cached receivers also need replacement donors. An extension covering missing positives alone would therefore be incomplete.

7.3 Historical claims awaiting dependency repair

The old sparse-feature attribution percentages, repair/ablation estimates, cross-dataset transfer, seed robustness, benign-only dictionary replication, and smaller-model mechanistic replication are not presented as corrected findings. Their dictionaries, ranking, negative pools, receivers, or donors can depend on the faulty labels. For the repaired-3B study in particular, correcting missing positive examples is insufficient: its label-selected negative sample must also be redrawn. Historical artifacts are preserved for exact reuse checks, not discarded; reuse requires the same complete intervention definition.

8 Interpretation and remaining work

The strongest corrected result is presently about the relationship between *evaluation population, score construction, and input dependence*. A model can achieve a high ROC under one comparison population, perform much less well on unseen users, and improve when only behavior targets contribute to its score. Those observations motivate mechanistic investigation; they do not already settle it.

The interpretability method remains useful: compare adaptation-induced states, fit a sparse approximation, select candidates on a declared discovery group, measure behavior-loss interventions against realized pertur-

bation controls, and report attribution with its location-versus-meaning limitation. Its methodological value depends on executing these checks and exposing negative or inconclusive outcomes. Existing sparse coding, causal contrasts, and feasibility mathematics do not become new mathematics merely through application to insider threats. The mathematical branch is retained as conditional work, with no new theorem or novelty claim asserted here.

Comparison-model dependencies. The separate classical architecture probe fitted with ten r4.2 official insiders and one r6.2 insider in its nominally normal training pool. For the local session autoencoders, an audit of actual saved bundles found contaminated training membership in 94/120 r4.2 bundles and 8/8 r6.2 bundles. All 120 r4.2 bundles have additional official insiders in their test groups. The 26 r4.2 bundles with clean training membership remain candidates for limited reuse; they do not clear the old benchmark as a whole. The older sequence-baseline harness shares the faulty fold construction and saves aggregate metrics, without the checkpoints and per-example predictions needed for score-only recovery. Refits must save scaler membership, checkpoints, predictions, and corrected fold definitions before architecture comparisons return.

Scope outside CERT. The CERT clock bug does not resolve separate TWOS or LANL issues. The early TWOS feature ranking had an independent defect; the corrected bounded held-out study remains inconclusive. LANL has an independent coupling between sampling and fold hashes. Neither is treated as a cleared replication here. Personality novelty associations on only five r6.2 insiders remain exploratory and are not used to explain the detector as a general law.

Next evidence needed. The next experiments must name the exact unsupported claim, corrected inputs, frozen endpoints, eligibility checks, output artifacts, and measured runtime before scaling. Priority goes to corrected causal selection and controls only where the retained paper claim needs them; comparator refits must first repair their training folds. Repeating every historical job is unnecessary, while relabeling every old output is insufficient. The 46-family recovery ledger records this distinction. This edition reports the verified subset and the remaining gaps; it is not a claim that the complete mechanistic program is finished or ready for submission.

9 Conclusion

Corrected labels preserve the main LM adapters but change several evaluated populations and invalidate some label-dependent experiment choices. Complete r4.2 behavior-target scoring improves ranking without changing the model or removing profile context. The interpretability question therefore remains: which internal pathways contribute to behavior predictions, and are their measured effects specific to incident-window activity? The current diagnostic pilot does not establish that specificity. A defensible account keeps this open question visible while retaining the verified score-level findings.

A Recovery provenance and scope

The machine-readable source is `paper/data/recovery_registry.json`. It records all 46 claim families, source paths and SHA-256 digests, exact metric selectors, completed scopes, limitations, and remaining work. Shared tables, values, and the mitigation figure are generated from that registry for all three editions. Source audits and the old manuscript texts are preserved in the repair branch history and in the external recovery archive. Historical figures and tables are not part of the corrected build.

Label validation. The official answer archive checksum was checked. Separate code enumerated the master-window calendar target and compared all metadata rows; it did not import the production label builder. Every named malicious event falls inside the corresponding user's extracted session interval under the recorded calendar clock. The r6.2 master includes a malformed CDE1846 start field, repaired using that named user's earliest recorded event. Master/detail discrepancies for ACM2278, CMP2946, and PLJ1771 are recorded rather than resolved by mixing target definitions. Corrected evaluation JSONLs preserve all non-label values.

Score recovery. The cached registry preserves corrected fold means, pooled metrics, paired contrasts, and fixed-sample results separately. Duplicate score identities must agree before deduplication. Each score view must have finite values and valid target counts; class losses must reconstruct total loss. The complete r4.2 extension follows an exact 256-row numerical bridge. The same-configuration class pass and the older full-population pass differ slightly numerically, which explains why the r4.2 full-score value in the mitigation table is 0.540 rather than the first table's 0.538. They must not be spliced into one causal comparison.

New CPU recovery. Pilot 2 and H6 retain their saved losses and historical feature selection. An independent arithmetic implementation reproduced 1,680 summaries and interval endpoints within 6×10^{-17}; pooled H6 values also reproduce the earlier independent audit. This checks aggregation, not the validity of the historical sampling protocol. Donor manifests were checked against the official insider list, independently of observed positive-day labels. The autoencoder audit reads actual saved fold membership and hashes each bundle, rather than inferring contamination from the manuscript's description.

Corrected causal pilot. The new r4.2 dictionary was fitted for 20 epochs on two million sampled token deltas, with layer width 4,096 and 8,192 sparse features, top-$k = 4$. Feature ranking used corrected discovery labels. The dictionary remains transductive; previously inspected confirmation users are not fresh holdouts. The full 158-pair run took approximately 45.9 minutes on an RTX 3070. Its original global norm gate failed at one control token. Separate diagnostic analyses preserve that failure, and the new conditional control-sampling feasibility probe is recorded as a distinct policy and artifact. The runtime measures this pilot, not the cost of all historical claim recovery.

Pending work and reusable artifacts. Raw sessions, corrected texts, main-adapter weights, and fixed-input responses remain useful. A contaminated scaler or detector requires refitting; a changed SAE dictionary requires downstream re-encoding, feature ranking, attribution, and required interventions. The repaired-3B negative sample also needs correction. The optional five-user personality-novelty association is deferred rather than promoted as an explanation. TWOS, LANL, and conditional geometric results retain their separate ledgers. The companion claim map and rerun specifications explain what each retained or deferred result would require.

B Relation to earlier methods

Anomaly detection supplies the one-class learning problem (Chandola et al., 2009); sequence and log models provide prior detector approaches (Du et al., 2017; Tuor et al., 2018; Guo et al., 2021). Shortcut learning and security-evaluation work motivate auditing the information behind apparent predictive success (Geirhos et al., 2020; Lapuschkin et al., 2019; Arp et al., 2022; Landauer et al., 2024). Sparse autoencoders provide candidate decompositions (Cunningham et al., 2023; Gao et al., 2024), while activation-patching methodology motivates matched interventions and careful interpretation (Zhang & Nanda, 2023; Heimersheim & Nanda, 2024). These established methods are the foundation of the audit. Applying them here does not itself establish a new sparse-coding algorithm, causal-identification theorem, or superior detector.

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